AI vs ML vs DL vs GEN AI

The Broad Field of AI:

The video defines **Artificial Intelligence** as the broad attempt to use computers to **simulate, match, or exceed human intelligence**, specifically the abilities to **learn, infer, and reason**. Historically, this began as a research project in the 1980s and 90s using programming languages like Lisp and Prolog to create **expert systems**.

Machine Learning (ML):

Machine learning represents a shift where the machine is not explicitly programmed but instead **learns by observing vast amounts of data**.

• **Pattern Recognition:** ML algorithms are highly effective at discovering patterns to make **predictions**.
• **Outlier Detection:** They are also used to spot **outliers**, which is particularly useful in fields like **cybersecurity** to identify users interacting with systems in atypical ways.

Deep Learning:

Deep learning is a subset of machine learning that uses **neural networks** to mimic the biological structures of the human brain.

• **Complexity:** It is called "deep" because it utilizes **multiple layers** of these networks.
• **Unpredictability:** Because these layers are so complex, it can be difficult for researchers to fully "decompose" or understand exactly why the system produces a specific result, making the technology somewhat **unpredictable**.

Generative AI (Gen AI):

The most recent advancement discussed is **Generative AI**, which is built upon **foundation models**.

• **Content Generation:** Unlike simple autocomplete features, technologies like **Large Language Models (LLMs)** can predict entire sentences, paragraphs, or documents.
• **Recombination as Creativity:** The source argues that even though Gen AI uses existing data, it is truly "generative" because it **recombines information into new permutations**, much like a composer creates new music using existing notes.
• **Applications and Risks:** This technology powers **chatbots** and **deep fakes**. While deep fakes can be used for entertainment or to assist people who have lost their voices, the video warns they can also be **abused**.

The Adoption Curve:

The video concludes by noting that while early AI adoption was slow and often felt "5 to 10 years away," the arrival of **foundation models and Generative AI** has caused adoption to "go straight to the moon," making AI a ubiquitous part of modern technology.


Artificial Intelligence Concepts
Artificial Intelligence (AI)
Definition
Simulate human intelligence
Learn, infer, and reason
History
Early research (LISP/Prolog)
Expert systems (1980s-90s)
Machine Learning (ML)
Core Concept
Learning without programming
Pattern discovery
Capabilities
Predictions
Spotting outliers
Use Cases
Cybersecurity
Deep Learning
Neural Networks
Simulate human brain
Multiple layers
Characteristics
Complex results
Hard to decompose
Generative AI
Foundation Models
Large Language Models (LLM)
Audio models
Video models
Outputs
Chatbots
Deep fakes
New content generation
Summarization